2. Random Variables

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FS26 MTH345: Deep Generative Models: A Statistical Perspective





Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- This lecture briefly reviews random variables, univariate and multivariate distributions, and the expectations of their functions.
- Useful materials include:
  - Mathematics for Machine Learning [MML] (Deisenroth et al., 2020)
  - A First Course in Probability [FCP] (Ross, 2020)
  - Mathematical Statistics and Data Analysis [MSDA] (Rice, 2007)
- See Chapter 6 in [MML] for details. We present content mostly following the textbooks, but also provide substantial unique and advanced material.
- Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

1. Random Variables

2. Discrete and Continuous Probability Distributions

3. Summary Statistics and Independence

4. Gaussian Distributions

Random Variables

- With the probability, we can describe probabilistic models of real-world phenomena involving randomness.
- In the two-coin toss example, suppose we are interested in the number of heads. Before the experiment, we do not know which $s \in S = \{HH, HT, TH, TT\}$ will occur, so the result is random (the probability describes this uncertainty).
- However, once the random experiment yields a specific s (e.g., HH), the number of heads is deterministically fixed (e.g., 2).

Random Variables

- **Random Variable**: A *random variable* (r.v.) is a function that maps each possible outcome in S to a quantity of interest in the *target space* $\mathcal{T}$.
- For example, $X : S = \{HH, HT, TH, TT\} \to \mathcal{T} = \{0, 1, 2\}$ defined by:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0 \tag{1}$$

  is the r.v. representing the number of heads.
- When the context is clear, it is common to omit the explicit outcome and write, e.g., $X = 2$ instead of $X(HH) = 2$.

Cumulative Distribution Function

- **(Cumulative) Distribution Function**: For a r.v. X,

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty \tag{2}$$

  is called the *cumulative distribution function (CDF)* or the *distribution function*.
- From now on, we omit the subscript denoting the random variable when the context is clear.

Cumulative Distribution Function

Exercise 1

Let X represent the number of heads obtained when tossing a coin twice. Find the CDF of X.

Cumulative Distribution Function

- The CDF F satisfies the following properties:
  1. F is a non-decreasing function.
  2. $F(\infty) := \lim_{x \to \infty} F(x) = 1$.
  3. $F(-\infty) := \lim_{x \to -\infty} F(x) = 0$.
  4. F is right-continuous.

  We denote the left limit at x by $F(x-)$, which satisfies $F(x-) = P(X < x)$.
- Conversely, for any function F satisfying the above four properties, there exists a probability P and a r.v. X such that $F_X = F$.

Discrete, Continuous, and Mixed Random Variables

- A r.v. is called *discrete* when the set of possible values of X is finite or countably infinite.
- For a discrete r.v., it is convenient to define a function to describe the size of the jumps in its CDF.
- **Probability Mass Function**: For a discrete r.v. X, the function p_X defined by

$$p_X(x) := P(X = x) = P(\{s : X(s) = x\}) \tag{3}$$

  is called the *probability mass function* (PMF).
- The PMF is related to the CDF by the jump size:

$$p(x) = P(X \leq x) - P(X < x) = F(x) - F(x-). \tag{4}$$

Discrete, Continuous, and Mixed Random Variables

Exercise 2

Let X represent the number of heads obtained when tossing a coin twice. Find the PMF of X.

Discrete, Continuous, and Mixed Random Variables

- A r.v. is called *continuous* when the set of possible values of X is uncountable.
- For a continuous r.v., it is convenient to define a function to describe the infinitesimal rate of increase of the CDF.
- **Probability Density Function**: For a continuous r.v. X, a function f_X satisfying the following conditions is called the *probability density function* (PDF):
  1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
  2. $P(a \leq X \leq b) = \int_a^b f_X(x)dx$ for any $a \leq b$.[1]

  Note that $P(X = a) = \int_a^a f_X(x)dx = 0$ for any a.

[1] This course uses f_X to distinguish density from probability, but it is also common to use p_X to denote the PDF.

Discrete, Continuous, and Mixed Random Variables

Exercise 3

A r.v. X is said to follow the *uniform distribution* over (a, b) for any $a < b$, denoted as $X \sim \text{Unif}(a, b)$, if its PDF is given by:

$$f(x; a, b) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Here, $(a, b)^\top$ parameterizes the PDF $f(\cdot; a, b)$. Find the CDF of X.

Discrete, Continuous, and Mixed Random Variables



Discrete, Continuous, and Mixed Random Variables

Exercise 4

Let X be a r.v. with parameter $\alpha > 0$, having the PDF:

$$f_X(x;\alpha) = \begin{cases} \alpha x^{\alpha-1} & \text{if } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

Compute $P(-1 < X < 1/2)$.

Discrete, Continuous, and Mixed Random Variables

- The PDF is related to the CDF by its derivative:

$$f(x) = \frac{d}{dx}F(x) = \lim_{\epsilon \to 0} \frac{F(x+\epsilon) - F(x)}{\epsilon} \tag{7}$$

(defined at points where F is differentiable). Consequently, for a small $\epsilon > 0$,

$$P(x - \epsilon/2 \leq X \leq x + \epsilon/2) \approx f(x)\epsilon. \tag{8}$$

- The probability of an interval is given by the difference in CDF values:

$$P(a < X \leq b) = F(b) - F(a). \tag{9}$$

Discrete, Continuous, and Mixed Random Variables

- **Types of Random Variables**:
  - X is *discrete* if F is a step function (piecewise constant).
  - X is *continuous* if F is a continuous function.
  - X is *mixed* if F is neither.
- In this course, for continuous r.v.s, we assume that F is differentiable, except possibly at countably many points.

Inverse Transform Sampling

- **Inverse Transform Sampling**: Let F_X be a strictly increasing CDF[2] and $U \sim \text{Unif}(0, 1)$. Then, the random variable defined by

$$Y = F_X^{-1}(U) \tag{10}$$

has the CDF F_X. That is, $Y \sim F_X$ since

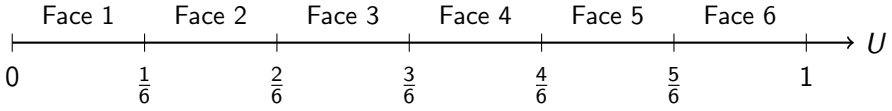
$$P(Y \leq y) = P(F_X^{-1}(U) \leq y) = P(U \leq F_X(y)) = F_X(y). \tag{11}$$

[2] The result extends to general CDFs by defining the generalized inverse $F_X^{-1}(u) = \inf\{x : F_X(x) \geq u\}$.

Inverse Transform Sampling

- For example, for a fair six-sided die, $\boldsymbol{p} = (1/6, \ldots, 1/6)^\top$ describes the distribution of outcomes.
- We can sample $U \sim \text{Unif}(0, 1)$ and determine the outcome as follows:

$$\text{Outcome} = \begin{cases} 1 & \text{if} \quad 0 \le U \le 1/6 \\ 2 & \text{if} \quad 1/6 < U \le 2/6 \\ 3 & \text{if} \quad 2/6 < U \le 3/6 \\ 4 & \text{if} \quad 3/6 < U \le 4/6 \\ 5 & \text{if} \quad 4/6 < U \le 5/6 \\ 6 & \text{if} \quad 5/6 < U \le 1 \end{cases}$$



Inverse Transform Sampling

Exercise 5

Let F_X be a strictly increasing CDF and $U \sim \text{Unif}(0, 1)$. Show that the random variable defined by $Y = F_X^{-1}(U)$ has CDF F_X.

Means and Variances

- **Expectation (Mean)**: The *expectation (mean)* of a discrete random variable X with probability mass function p_X is:

$$\mu_X = \mathrm{E}[X] = \sum_x x p_X(x). \tag{12}$$

Correspondingly, the expectation of a continuous random variable X with probability density function f_X is:

$$\mu_X = \mathrm{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx. \tag{13}$$

Exercise 6

Let $X \sim \mathrm{Unif}(a, b)$. Show that $\mathrm{E}[X] = (a + b)/2$.

Means and Variances

- **Law of the Unconscious Statistician**: If X is a discrete random variable, then for any real-valued function g, $\mathrm{E}[g(X)] = \sum_x g(x)p_X(x)$. Similarly, for a continuous random variable X, $\mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)\,dx$.[3]
- **Variance**: The *variance* of X is defined as the expected squared Euclidean distance between X and its mean:

$$\sigma_X^2 = \mathrm{Var}[X] = \mathrm{E}[(X - \mathrm{E}[X])^2]. \tag{14}$$

Exercise 7

Let $X \sim \mathrm{Unif}(a, b)$. Show that $\mathrm{Var}[X] = (b-a)^2/12$.

[3] See [FCP] Proposition 4.1 for details.

Means and Variances

Exercise 8

Show that:

1. For any real numbers a and b, $\mathrm{E}[aX + b] = a\,\mathrm{E}[X] + b$.
2. For any real numbers a and b, $\mathrm{Var}[aX + b] = a^2\,\mathrm{Var}[X]$.
3. $\mathrm{Var}[X] = \mathrm{E}[X^2] - \big(\mathrm{E}[X]\big)^2$.
4. For a discrete random variable X, $\mathrm{Var}[X] = 0$ if and only if X is deterministic, i.e., $P(X = \mu_X) = 1$.[a]

[a] This result holds in general for continuous and mixed random variables as well.

Means and Variances



Means and Variances

- **Standard Deviation**: The *standard deviation* of X is defined as $\sigma_X = \sqrt{\mathrm{Var}[X]}$.
- For a given X with $\sigma_X > 0$, its standardized random variable Z (called Z-*score*) is defined as $Z = (X - \mu_X)/\sigma_X$.

Exercise 9

Let X and Y denote exam scores in Statistics and Linear Algebra, respectively. Given $\mu_X = 50$, $\sigma_X = 10$, $\mu_Y = 60$, and $\sigma_Y = 5$, compare the Z-scores corresponding to a score of 100 in both courses.

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Univariate Discrete Random Variables

- Common univariate discrete distributions include the Bernoulli and binomial distributions.
- **Bernoulli Random Variable**: A random variable X is said to follow the *Bernoulli distribution*, denoted as $X \sim \text{Ber}(\mu)$ where $0 \leq \mu \leq 1$, if its PMF is given by:

$$p_X(x;\mu) = \begin{cases} \mu & \text{if } x = 1, \\ 1-\mu & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases} \tag{15}$$

- The PMF of $X \sim \text{Ber}(\mu)$ can be summarized in the following table:

x	0	1
$p_X(x;\mu)$	$1-\mu$	μ

Univariate Discrete Random Variables

Exercise 10

Let $X \sim \text{Ber}(\mu)$. Show that $\text{E}[X] = \mu$ and $\text{Var}[X] = \mu(1 - \mu)$.

Univariate Discrete Random Variables

Exercise 11

Consider a random experiment consisting of n trials, where each trial results success with probability μ or a failure with probability $1 - \mu$. Assume that the trials do not affect each other. Let X be the number of successes. Find the PMF of X.

Univariate Discrete Random Variables

- **Binomial Random Variable**: A random variable X is said to follow the *binomial distribution*, denoted as $X \sim \mathrm{Bin}(n, \mu)$, where $n \in \mathbb{N}$ and $0 \leq \mu \leq 1$, if its PMF is given by:

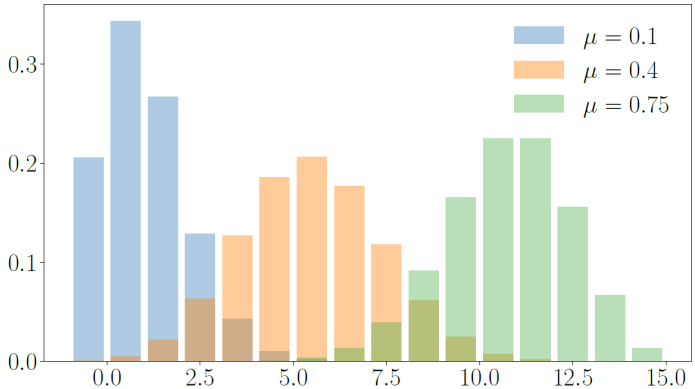
$$p_X(x; n, \mu) = \begin{cases} \binom{n}{x} \mu^x (1-\mu)^{n-x} & \text{if } x \in \{0, 1, \ldots, n\}, \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

  Here, $\binom{n}{x}$ denotes the binomial coefficient, defined as $\frac{n!}{x!(n-x)!}$, which represents the number of distinct ways to choose x objects from n distinguishable objects.

- The PMF of $X \sim \mathrm{Bin}(n, \mu)$ can be summarized in the following table:

x	0	1	...	n
$p_X(x; n, \mu)$	$(1-\mu)^n$	$n\mu(1-\mu)^{n-1}$	...	μ^n

Univariate Discrete Random Variables



[MML] Fig. 6.10: The PMFs of $\mathrm{Bin}(n, \mu)$ when $(n, \mu) = (15, 0.1)$, $(15, 0.4)$, and $(15, 0.75)$.

Univariate Discrete Random Variables

Exercise 12

Let $X \sim \text{Bin}(n, \mu)$. Show that $\text{E}[X] = n\mu$ and $\text{Var}[X] = n\mu(1 - \mu)$.

Univariate Continuous Random Variables

- Common univariate continuous distributions include the uniform and Gaussian distributions.
- **Gaussian Random Variable**: A random variable X is said to follow the *Gaussian (or normal) distribution*[4] with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted by $X \sim \mathcal{N}(\mu, \sigma^2)$, if its PDF is given by:

$$f_X(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{17}$$

- The PDF of the *standard normal random variable* $Z \sim \mathcal{N}(0, 1)$ is denoted by:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right). \tag{18}$$

[4] An alternative definition is $X = \mu + \sigma Z$, where Z is a random variable with PDF $\phi(z)$. This course derives this as a property in Exercise 14.

Univariate Continuous Random Variables

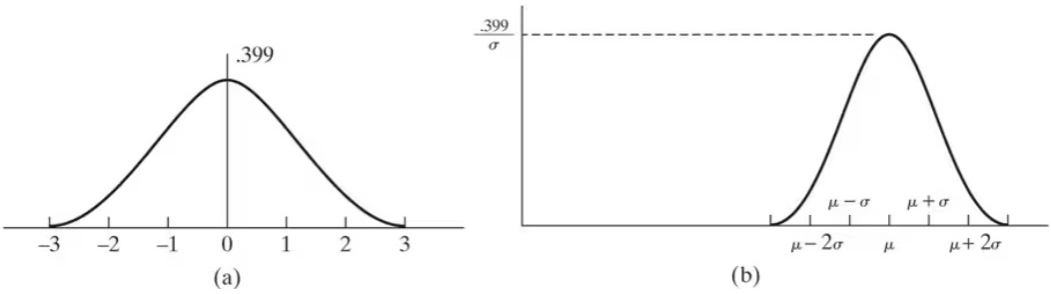


Fig. 5.5 [FCP]: Normal density function: (a) $\mu = 0, \sigma = 1$; (b) arbitrary μ, σ^2.

- We denote the PDF and CDF of the *standard Gaussian distribution* $\mathcal{N}(0, 1)$ by $\phi(z)$ and $\Phi(z)$, respectively.

Univariate Continuous Random Variables

Exercise 13

Show that the PDF of $X \sim \mathcal{N}(\mu, \sigma^2)$ integrates to 1, i.e., for any μ and σ,

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = 1. \tag{19}$$

Univariate Continuous Random Variables

Univariate Continuous Random Variables

Exercise 14

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Show that $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$ for any real constants a and b.

Multivariate Random Variables

- A *random vector* is a vector whose components are random variables.
- For example, when tossing a coin twice, the mapping $\boldsymbol{X} := (X_1, X_2)^\top$ defined on the sample space $S = \{HH, HT, TH, TT\}$ by

$$\boldsymbol{X}(HH) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \boldsymbol{X}(HT) = \boldsymbol{X}(TH) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \boldsymbol{X}(TT) = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \tag{20}$$

  is a random vector representing the counts of heads and tails, respectively.

Multivariate Random Variables

- **Joint (Cumulative) Distribution Function**: For a random vector $\boldsymbol{X} := (X_1, \ldots, X_d)^\top$, its joint CDF is given by:

$$F_{\boldsymbol{X}}(\boldsymbol{x}) = P(X_1 \leq x_1, \ldots, X_d \leq x_d) \tag{21}$$

where $\boldsymbol{x} := (x_1, \ldots, x_d)^\top \in \mathbb{R}^d$.

Exercise 15

When we toss a coin twice, let $\boldsymbol{X} := (X_1, X_2)^\top$ where X_1 and X_2 are numbers of heads and tails, respectively. Calculate $F_{\boldsymbol{X}}((1,0)^\top)$, $F_{\boldsymbol{X}}((1,1)^\top)$, and $F_{\boldsymbol{X}}((2,0)^\top)$.

Multivariate Random Variables

- **Joint Probability Mass Function**: For a jointly discrete random vector $\boldsymbol{X}$, its joint PMF $p_{\boldsymbol{X}}$ is given by:

$$p_{\boldsymbol{X}}(\boldsymbol{x}) := P(\boldsymbol{X} = \boldsymbol{x}) = P(\{s : X_1(s) = x_1, X_2(s) = x_2, \ldots, X_d(s) = x_d\}). \tag{22}$$

Exercise 16

When we toss a coin twice, let $\boldsymbol{X} := (X_1, X_2)^\top$ where X_1 and X_2 are numbers of heads and tails, respectively. Calculate $p_{\boldsymbol{X}}((1,0)^\top)$, $p_{\boldsymbol{X}}((1,1)^\top)$, and $p_{\boldsymbol{X}}((2,0)^\top)$.

Multivariate Random Variables

- **Example 6.1a from [FCP]**: Suppose that three balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. Let X and Y denote the number of red and white balls chosen, respectively. Then, the joint PMF of X and Y is given by:

$$p_{X,Y}(i,j) = \frac{\binom{3}{i}\binom{4}{j}\binom{5}{3-i-j}}{\binom{12}{3}} \tag{23}$$

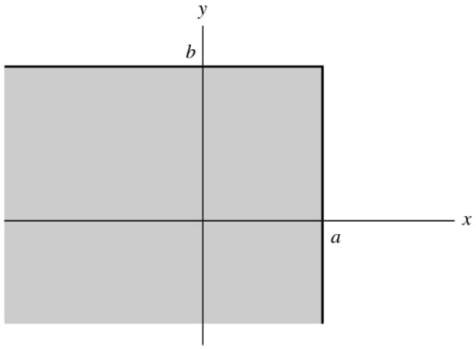
for non-negative integers i and j such that $i + j \leq 3$.

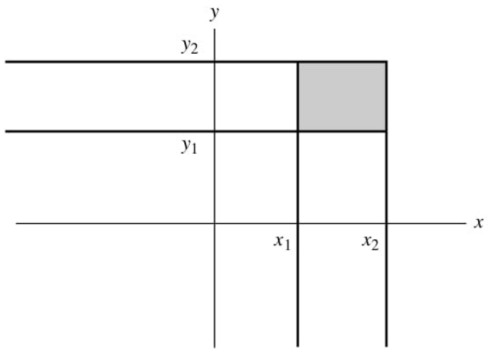
Multivariate Random Variables

i \ j	0	1	2	3	Row sum $= P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum $= P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

[FCP] Tab. 6.1: $p_{X,Y}(i,j) = P(X = i, Y = j)$.

Multivariate Random Variables





Figs 3.1 and 3.2 from [MSDA]:
$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1)$

- The joint PMF can also be recovered directly from the joint CDF:

$$p(x, y) = F(x, y) - F(x-, y) - F(x, y-) + F(x-, y-). \quad (24)$$

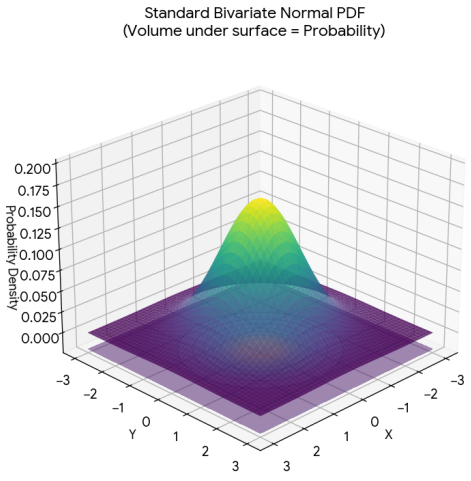
Multivariate Random Variables

- **Joint Probability Density Function**: For a jointly continuous random vector $\boldsymbol{X}$, its joint PDF $f_{\boldsymbol{X}}$ satisfies:
  1. $f_{\boldsymbol{X}}(\boldsymbol{x}) \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^d$.
  2. $P(\boldsymbol{X} \in (a_1, b_1) \times \cdots \times (a_d, b_d)) = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f_{\boldsymbol{X}}(x_1, \ldots, x_d)\, dx_d \ldots dx_1$.
- For example, let $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ be the PDFs of X_1 and X_2, respectively. Then, the product

$$f_{X_1}(x_1) f_{X_2}(x_2), \tag{25}$$

which is a function of $(x_1, x_2)^\top \in \mathbb{R}^2$, satisfies the conditions for a joint PDF.

Multivariate Random Variables



- When f_{X_1} and f_{X_2} are the PDFs of $\mathcal{N}(0, 1^2)$,

$$(x_1, x_2)^\top \mapsto f_{X_1}(x_1) f_{X_2}(x_2) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}\left(x_1^2 + x_2^2\right)\right) \tag{26}$$

defines a joint PDF of $\boldsymbol{X} = (X_1, X_2)^\top$.

Multivariate Random Variables

- The joint PDF is related to the joint CDF by its derivative:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\partial^d}{\partial x_1 \cdots \partial x_d} F_{\boldsymbol{X}}(\boldsymbol{x}). \tag{27}$$

Consequently, for a small $\epsilon_j > 0$,

$$P(x_1 - \epsilon_1/2 \leq X_1 \leq x_1 + \epsilon_1/2, \ldots, x_d - \epsilon_d/2 \leq X_d \leq x_d + \epsilon_d/2) \approx f_{\boldsymbol{X}}(\boldsymbol{x})\epsilon_1 \cdots \epsilon_d. \tag{28}$$

Multivariate Random Variables

- **Marginal Probability Mass Function**: Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be random vectors with joint PMF $p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})$. Then, the *marginal PMF* $p_{\boldsymbol{X}}(\boldsymbol{x})$ is given by:

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \sum_{\boldsymbol{y}\in\mathcal{Y}} p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) \tag{29}$$

- **Marginal Probability Density Function**: Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be random vectors with joint PDF $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})$. Then, the *marginal PDF* of $\boldsymbol{X}$, denoted $f_{\boldsymbol{X}}(\boldsymbol{x})$, is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \int_{\mathbb{R}^{d_{\boldsymbol{Y}}}} f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})\, d\boldsymbol{y}. \tag{30}$$

Multivariate Random Variables

Exercise 17

When we toss a coin twice, let $\boldsymbol{X} := (X_1, X_2)^\top$ where X_1 and X_2 are numbers of heads and tails, respectively. Find the marginal PMF of X_1.

Multivariate Random Variables

Exercise 18

Let $\boldsymbol{X} := (X_1, X_2)^\top$ be a continuous random vector whose joint PDF is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}\left(x_1^2 + x_2^2\right)\right). \tag{31}$$

Find the marginal PDF of X_1.

Multivariate Random Variables

- **Multinomial Random Vector**: A random vector $\boldsymbol{X}$ is said to follow the *multinomial distribution*, denoted by $\boldsymbol{X} \sim \mathrm{Multin}(n, \boldsymbol{\mu})$, where $n \in \mathbb{N}$, $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_d)^\top$, $\mu_j \geq 0$, and $\sum_{j=1}^{d} \mu_j = 1$, if its PMF is given by:

$$p(\boldsymbol{x}; n, \boldsymbol{\mu}) = \begin{cases} \binom{n}{x_1, x_2, \ldots, x_d} \mu_1^{x_1} \mu_2^{x_2} \cdots \mu_d^{x_d} & \text{if } x_j \in \{0, 1, \ldots, n\} \text{ and } \sum_{j=1}^{d} x_j = n, \\ 0 & \text{otherwise.} \end{cases} \tag{32}$$

- The multivariate Gaussian distribution is one of the most widely used continuous multivariate distributions. We will discuss it in detail in a later subsection.

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Means and Covariances

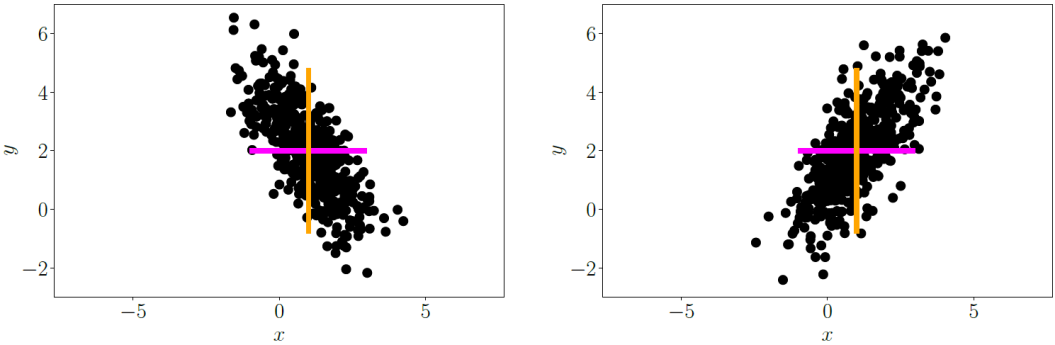
- For a multivariate r.v. $\boldsymbol{X}(\cdot) := (X_1(\cdot), \ldots, X_d(\cdot))^\top$, its expectation is defined as:

$$\boldsymbol{\mu_X} = \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}] = \left(\mathrm{E}_{X_1}[X_1], \ldots, \mathrm{E}_{X_d}[X_d]\right)^\top \in \mathbb{R}^d. \tag{33}$$

- The expectation operator is linear. For any real constants a and b, and any real-valued functions g and h defined on the support of $\boldsymbol{X}$,

$$\mathrm{E}[ag(\boldsymbol{X}) + bh(\boldsymbol{X})] = a\,\mathrm{E}[g(\boldsymbol{X})] + b\,\mathrm{E}[h(\boldsymbol{X})]. \tag{34}$$

Means and Covariances



[MML] Fig. 6.5: Scatter plots of two observed random variables X and Y.

- **Covariance**: The *covariance* between two univariate r.v.s X and Y is given by:

$$\sigma_{X,Y} = \mathrm{Cov}_{X,Y}[X, Y] = \mathrm{E}_{X,Y}\left[(X - \mathrm{E}_X[X])(Y - \mathrm{E}_Y[Y])\right]. \tag{35}$$

Means and Covariances

Exercise 19

Show that:

1. $\mathrm{Cov}[aX + b, cY + d] = ac\,\mathrm{Cov}[X, Y]$.
2. $\mathrm{Cov}[X, Y] = \mathrm{E}[XY] - \mathrm{E}[X]\,\mathrm{E}[Y]$.
3. $\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}[X, Y]$
4. $\mathrm{Var}[X - Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] - 2\,\mathrm{Cov}[X, Y]$

Means and Covariances

Exercise 20

For an event A, let I_A denote the *indicator random variable* of A, defined by:

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases} \tag{36}$$

For any two events A and B with $P(A) > 0$, show that:

$$\text{Cov}[I_A, I_B] = P(A)\big(P(B \mid A) - P(B)\big). \tag{37}$$

Means and Covariances

- **Correlation**: Let X and Y be univariate random variables having positive variances. The *correlation* between X and Y is given by:

$$\rho_{X,Y} = \text{Corr}_{X,Y}[X, Y] = \frac{\text{Cov}_{X,Y}[X, Y]}{\sqrt{\text{Var}_X[X]\,\text{Var}_Y[Y]}}. \tag{38}$$

- Again, the *standardized variable* (or *Z-score*) of X is defined as:

$$Z_X = \frac{X - \text{E}[X]}{\sqrt{\text{Var}[X]}}. \tag{39}$$

The correlation is equal to the covariance between Z-scores:

$$\text{Corr}[X, Y] = \text{Cov}\left[\frac{X - \text{E}[X]}{\sqrt{\text{Var}[X]}}, \frac{Y - \text{E}[Y]}{\sqrt{\text{Var}[Y]}}\right]. \tag{40}$$

Means and Covariances

Exercise 21

Let Z_X and Z_Y be the Z-scores of random variables X and Y, respectively.

1. Show that:

$$\text{Var}[Z_X + Z_Y] = 2(1 + \rho_{X,Y}) \quad \text{and} \quad \text{Var}[Z_X - Z_Y] = 2(1 - \rho_{X,Y}).$$

2. Hence, conclude that $-1 \leq \rho_{X,Y} \leq 1$.

Means and Covariances

- **Covariance Matrix**: The variance of a r.v. $\boldsymbol{X}$ is defined as:

$$\Sigma_{\boldsymbol{X}} = \mathrm{Var}_{\boldsymbol{X}}[\boldsymbol{X}] = \mathrm{E}_{\boldsymbol{X}}[(\boldsymbol{X} - \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}])(\boldsymbol{X} - \mathrm{E}_{\boldsymbol{X}}[\boldsymbol{X}])^{\top}] \tag{41}$$

  This matrix is referred to as the *covariance matrix* of $\boldsymbol{X}$.

- For a $d_{\boldsymbol{X}}$-dimensional random vector $\boldsymbol{X}$ and a $d_{\boldsymbol{Y}}$-dimensional random vector $\boldsymbol{Y}$, the cross-covariance matrix is defined as:

$$\boldsymbol{\Sigma}_{\boldsymbol{X},\boldsymbol{Y}} = \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] = \mathrm{E}\left[(\boldsymbol{X} - \mathrm{E}[\boldsymbol{X}])(\boldsymbol{Y} - \mathrm{E}[\boldsymbol{Y}])^{\top}\right] \in \mathbb{R}^{d_{\boldsymbol{X}} \times d_{\boldsymbol{Y}}}. \tag{42}$$

Means and Covariances

Exercise 22

Let $\boldsymbol{Z} := (Z_1, Z_2)^\top$ be a random vector satisfying $\mathrm{E}[\boldsymbol{Z}] = 0$ and $\mathrm{Cov}[\boldsymbol{Z}] = \boldsymbol{I}_2$. Define

$$X_1 = \mu_1 + \sigma_1 Z_1,$$
$$X_2 = \mu_2 + \sigma_2 \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right),$$

where $(\mu_1, \mu_2)^\top \in \mathbb{R}^2$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $\rho \in [-1, 1]$. Show that:

$$\mathrm{Corr}[X_1, X_2] = \rho. \tag{43}$$

Means and Covariances

Exercise 23

Let $\boldsymbol{X} \in \mathbb{R}^{d_X}$ and $\boldsymbol{Y} \in \mathbb{R}^{d_Y}$ be random vectors. Show that:

1. For any constant matrix A and constant vector $\boldsymbol{b}$ of compatible dimensions,

$$\mathrm{E}[A\boldsymbol{X} + \boldsymbol{b}] = A\,\mathrm{E}[\boldsymbol{X}] + \boldsymbol{b}. \tag{44}$$

2. For any constant matrices A and B, and constant vectors $\boldsymbol{c}$ and $\boldsymbol{d}$ of compatible dimensions,

$$\mathrm{Cov}[A\boldsymbol{X} + \boldsymbol{c}, B\boldsymbol{Y} + \boldsymbol{d}] = A\,\mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]B^{\top}. \tag{45}$$

Means and Covariances

Exercise 24

Let $\boldsymbol{X} \in \mathbb{R}^{d_{\boldsymbol{X}}}$ and $\boldsymbol{Y} \in \mathbb{R}^{d_{\boldsymbol{Y}}}$ be random vectors. Show that:

1. $\big(\mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]\big)_{jk} = \mathrm{Cov}[X_j, Y_k]$ for all $j \in \{1, \ldots, d_{\boldsymbol{X}}\}$ and $k \in \{1, \ldots, d_{\boldsymbol{Y}}\}$.
2. $\mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] = \mathrm{E}[\boldsymbol{X}\boldsymbol{Y}^\top] - \mathrm{E}[\boldsymbol{X}]\,\mathrm{E}[\boldsymbol{Y}]^\top$.
3. When $d_{\boldsymbol{X}} = d_{\boldsymbol{Y}}$,
   $\mathrm{Var}[\boldsymbol{X} + \boldsymbol{Y}] = \mathrm{Var}[\boldsymbol{X}] + \mathrm{Var}[\boldsymbol{Y}] + \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] + \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]^T$
4. When $d_{\boldsymbol{X}} = d_{\boldsymbol{Y}}$,
   $\mathrm{Var}[\boldsymbol{X} - \boldsymbol{Y}] = \mathrm{Var}[\boldsymbol{X}] + \mathrm{Var}[\boldsymbol{Y}] - \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] - \mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}]^T$

Means and Covariances



Means and Covariances

- **Sample Mean (Empirical Mean)**: Let $x_1, \ldots, x_n$ be n observations of a random variable X. Then, the *sample mean* of X is defined as $\bar{x} = n^{-1} \sum_{i=1}^{n} x_i$.
- **Sample Variance**: Let $x_1, \ldots, x_n$ be n observations of a random variable X. Then, the *sample variance* of X is defined as $s_{X,X} = s_X^2 = (n-1)^{-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$.

Exercise 25

Let $x_1, \ldots, x_n$ be n observations of a random variable X. Show that

$$s_X^2 = \frac{1}{2\binom{n}{2}} \sum_{1 \le i < j \le n} (x_i - x_j)^2. \tag{46}$$

Means and Covariances

- **Sample Covariance**: Let $(x_1, y_1)^\top, \ldots, (x_n, y_n)^\top$ be n observations of a random vector $(X, Y)^\top$. Then, the *sample covariance* between X and Y is defined as $s_{X,Y} = (n-1)^{-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$.
- **Sample Correlation**: Let $(x_1, y_1)^\top, \ldots, (x_n, y_n)^\top$ be n observations of a random vector $(X, Y)^\top$. Assume $s_X > 0$ and $s_Y > 0$. Then, the *sample correlation* is defined as $r_{X,Y} = s_{X,Y}/(s_X s_Y)$.
- In a similar manner, we can define multivariate versions of the sample mean and covariance matrix.

Statistical Independence

- **Statistical Independence**: Two random variables X and Y are said to be *independent* if, for any two intervals I_X and I_Y of real numbers,

$$P(X \in I_X, Y \in I_Y) = P(X \in I_X)P(Y \in I_Y). \tag{47}$$

  That is, the events $X^{-1}(I_X)$ and $Y^{-1}(I_Y)$ are independent.

- We write $X \perp\!\!\!\perp Y$ to denote that X is independent of Y.
- Random variables $X_1, \ldots, X_n$ are said to be *(mutually) independent* if, for any intervals $I_1, \ldots, I_n \subset \mathbb{R}$,

$$P(X_1 \in I_1, \ldots, X_n \in I_n) = \prod_{i=1}^{n} P(X_i \in I_i). \tag{48}$$

  That is, the events $X_1^{-1}(I_1), \ldots, X_n^{-1}(I_n)$ are mutually independent.

Statistical Independence

- Random vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$ are independent if and only if either of the following equivalent conditions holds:
  1. $F_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = F_{\boldsymbol{X}}(\boldsymbol{x})F_{\boldsymbol{Y}}(\boldsymbol{y})$
  2. $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = f_{\boldsymbol{X}}(\boldsymbol{x})f_{\boldsymbol{Y}}(\boldsymbol{y})$ (or $p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = p_{\boldsymbol{X}}(\boldsymbol{x})p_{\boldsymbol{Y}}(\boldsymbol{y})$ in the discrete case).
- An infinite collection of random vectors is said to be independent if every finite sub-collection is independent.

Statistical Independence

Exercise 26

Show that random variables X and Y are independent if and only if their joint PDF is factorizable, i.e.,

$$f_{X,Y}(x, y) = g(x)h(y) \tag{49}$$

for some functions g and h.

Statistical Independence

Exercise 27

If $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, then for any functions g and h, $g(\boldsymbol{X}) \perp\!\!\!\perp h(\boldsymbol{Y})$.

Statistical Independence

Exercise 28

([FCP] Example 2f) Let the joint density function of X and Y be

$$f_{X,Y}(x,y) = 24xy, \quad 0 < x < 1,\ 0 < y < 1,\ 0 < x + y < 1, \tag{50}$$

and 0 otherwise. Find marginal PDFs and compare their product with $f_{X,Y}(x,y)$.

Statistical Independence

- For any two independent random vectors $\boldsymbol{X} \in \mathbb{R}^d$ and $\boldsymbol{Y} \in \mathbb{R}^d$:

$$\mathrm{Cov}[\boldsymbol{X}, \boldsymbol{Y}] = 0. \tag{51}$$

  However, the converse does not hold in general.

- **Example 6.5 from [MML]**: Consider a non-constant random variable X satisfying $\mathrm{E}[X] = 0$ and $\mathrm{E}[X^3] = 0$. Let $Y = X^2$. Then, $\mathrm{Cov}[X, Y] = 0$, but X and Y are not independent.

Outline

1. Random Variables

2. Discrete and Continuous Probability Distributions

3. Summary Statistics and Independence

4. Gaussian Distributions

Motivation for Gaussian Distributions

- In the univariate case, the PDF of the Gaussian distribution with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted by $\mathcal{N}(\mu, \sigma^2)$, is defined as:

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{52}$$

- We review two motivations for Gaussian distributions:
  1. Limit behavior of the sample mean (*Central Limit Theorem*)
  2. Rotation invariance with independent components (*Maxwell's Theorem*)

Motivation for Gaussian Distributions

- For the first motivation, suppose we want to check if a coin is fair. After flipping the coin 100 times, we observe 40 heads (relative frequency 0.4).
- Can we conclude that the true probability of heads is less than 0.5? Although $0.4 < 0.5$, this could simply be due to randomness.
- A typical approach in statistics is to calculate the probability of observing a result as extreme as (or more extreme than) the observation.
- Specifically, let $X_1, \ldots, X_n$ be i.i.d. samples from Ber(0.5). The relative frequency is the sample mean $\bar{X} = n^{-1} \sum_{i=1}^{n} X_i$. We need to understand the distributional behavior of $\bar{X}$. For example, when $n = 100$, the probability of observing $\bar{X} \leq 0.4$ is:

$$P(\bar{X} \leq 0.4) = \sum_{i=0}^{40} \binom{100}{i} 0.5^i (1 - 0.5)^{100-i} \approx 1.76\%. \tag{53}$$

Motivation for Gaussian Distributions

- Surprisingly, for any population with finite variance, the standardized sample mean converges to a unique distribution. It is natural to define this limit distribution as a standard reference, which is the *standard Gaussian distribution* $\mathcal{N}(0,1)$.[5]
- Let $Z_{\bar{X}}$ be the Z-score of the sample mean where the population mean is μ and variance is σ^2:

$$Z_{\bar{X}} = \frac{\bar{X} - \mathrm{E}[\bar{X}]}{\sqrt{\mathrm{Var}[\bar{X}]}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}. \tag{54}$$

As $n \to \infty$, the sequence of PDFs of $Z_{\bar{X}}$ converges to a limit PDF that is invariant to the original distribution of X_i:

$$f_{Z_{\bar{X}}}(z) \xrightarrow{n\to\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \quad \text{for all } z \in \mathbb{R}. \tag{55}$$

Here, the limit PDF is the density of the standard Gaussian, denoted as $\phi(z)$.

[5] This result is known as the **Central Limit Theorem.**

Motivation for Gaussian Distributions

- Thus, when n is sufficiently large, $\bar{X}$ is approximately:

$$\bar{X} \approx \mu + \frac{\sigma}{\sqrt{n}} Z \tag{56}$$

where $Z \sim \mathcal{N}\left(0, 1^2\right)$. The PDF of this approximation is given by:

$$f_{\bar{X}}(\bar{x}) \approx \frac{1}{\sqrt{2\pi(\sigma^2/n)}} \exp\left(-\frac{(\bar{x}-\mu)^2}{2(\sigma^2/n)}\right). \tag{57}$$

- In this context, we parametrize the general class of PDFs as:

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{58}$$

which includes the approximated distribution of $\bar{X}$ as $\mathcal{N}(\mu, \sigma^2/n)$.

Motivation for Gaussian Distributions

- This motivation naturally extends to the multivariate case.
- Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \in \mathbb{R}^d$ be n independent and identically distributed (i.i.d.) random vectors with population mean $\mathrm{E}[\boldsymbol{X}] = \boldsymbol{\mu}$ and covariance $\mathrm{Var}[\boldsymbol{X}] = \Sigma$.
- By the Multivariate Central Limit Theorem, for sufficiently large n, $\bar{\boldsymbol{X}}$ is approximately distributed as:

$$\bar{\boldsymbol{X}} \approx \boldsymbol{\mu} + \frac{1}{\sqrt{n}} \boldsymbol{\Sigma}^{1/2} \boldsymbol{Z}, \tag{59}$$

  where $\boldsymbol{Z} := (Z_1, \ldots, Z_d)^\top$ consists of d independent standard normal random variables.
- To derive the joint PDF of the approximating Gaussian random vector

$$\boldsymbol{\mu} + \frac{1}{\sqrt{n}} \boldsymbol{\Sigma}^{1/2} \boldsymbol{Z},$$

  we first review the change of variables formula.

Change of Variables Formula

- Let $\boldsymbol{X}$ be a continuous random vector with PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$ supported on a region $T_X \subseteq \mathbb{R}^d$. Let $\boldsymbol{Y} = g(\boldsymbol{X})$, where g is a one-to-one, differentiable transformation from T_X onto T_Y. Then, $\boldsymbol{Y}$ has the PDF:

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \begin{cases} f_{\boldsymbol{X}}(g^{-1}(\boldsymbol{y})) \left| \det \frac{\partial g^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}} \right|, & \text{if } \boldsymbol{y} \in T_Y, \\ 0, & \text{otherwise.} \end{cases} \tag{60}$$

Here, $\left| \det \frac{\partial g^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}} \right|$ denotes the absolute value of the determinant of the Jacobian matrix.

Change of Variables Formula

- When $g : \mathbb{R}^d \to \mathbb{R}^d$ is differentiable and invertible on T_X, and $g^{-1}(\boldsymbol{y}) \in T_X$, denoting the hypercube $C_\epsilon(\boldsymbol{y}) = [\boldsymbol{y} - \frac{\epsilon}{2}\mathbf{1}, \boldsymbol{y} + \frac{\epsilon}{2}\mathbf{1}]$ where $\mathbf{1} := (1, \ldots, 1)^\top$:

$$
\begin{aligned}
f_{\boldsymbol{Y}}(\boldsymbol{y}) &= \lim_{\epsilon \to 0^+} \frac{P(\boldsymbol{Y} \in C_\epsilon(\boldsymbol{y}))}{\epsilon^d} \\
&= \lim_{\epsilon \to 0^+} \frac{P(\boldsymbol{X} \in g^{-1}(C_\epsilon(\boldsymbol{y})))}{\epsilon^d} \\
&= \lim_{\epsilon \to 0^+} \frac{P(\boldsymbol{X} \in g^{-1}(C_\epsilon(\boldsymbol{y})))}{\mathrm{Vol}(g^{-1}(C_\epsilon(\boldsymbol{y})))} \frac{\mathrm{Vol}(g^{-1}(C_\epsilon(\boldsymbol{y})))}{\epsilon^d} \\
&= f_{\boldsymbol{X}}(g^{-1}(\boldsymbol{y})) \left| \det \frac{\partial g^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}} \right|.
\end{aligned}
\tag{61}
$$

Change of Variables Formula

- **Change of Variables Formula**: Suppose there exists a partition of T_X such that $T_X = \bigcup_{i=1}^{k} T_{X,i}$ (where $T_{X,i} \cap T_{X,j} = \emptyset$ for $i \neq j$), and the restriction of g to $T_{X,i}$, denoted by g_i, is a one-to-one differentiable mapping. Then, $\boldsymbol{Y}$ has the PDF:

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \sum_{i=1}^{k} f_{\boldsymbol{X}}(g_i^{-1}(\boldsymbol{y})) \left| \det \frac{\partial g_i^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}} \right|, \tag{62}$$

  where the sum is over indices i such that $\boldsymbol{y}$ is in the image of g_i.
- The change of variables formula is fundamental in defining multivariate distributions.

Motivation for Gaussian Distributions

Exercise 29

Let $\boldsymbol{X} = \boldsymbol{\mu} + \Sigma^{1/2}\boldsymbol{Z}$ where $\boldsymbol{Z} := (Z_1, \ldots, Z_d)^\top$ consists of d independent standard normal random variables. Use the change of variables formula to show that:

$$f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = (2\pi)^{-d/2}|\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\right). \tag{63}$$

Motivation for Gaussian Distributions

- **Multivariate Gaussian Distribution**: A random vector $\boldsymbol{X} \in \mathbb{R}^d$ is said to follow the *multivariate Gaussian distribution* with parameters $\boldsymbol{\mu} \in \mathbb{R}^d$ and positive definite matrix $\Sigma \in \mathbb{R}^{d \times d}$, denoted by $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if its joint PDF is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right). \tag{64}$$

- Here, the exponent is proportional to the negative of the squared *Mahalanobis distance*. For given points $\boldsymbol{x}$ and $\boldsymbol{y}$ and a positive definite matrix A, the Mahalanobis distance between $\boldsymbol{x}$ and $\boldsymbol{y}$ with respect to A is defined as:

$$d_M(\boldsymbol{x}, \boldsymbol{y}; A) = \sqrt{(\boldsymbol{x} - \boldsymbol{y})^T A^{-1} (\boldsymbol{x} - \boldsymbol{y})}. \tag{65}$$

That is, as $d_M^2(\boldsymbol{x}, \boldsymbol{\mu}; \Sigma)$ increases, the probability density decreases exponentially.

Motivation for Gaussian Distributions

- The following is also a common definition of the multivariate Gaussian distribution: A random vector $\boldsymbol{X}$ follows $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ if and only if there exist a matrix $A \in \mathbb{R}^{d \times k}$ and a random vector $\boldsymbol{Z} \sim \mathcal{N}(0, I_k)$ such that:

$$\boldsymbol{X} = \boldsymbol{\mu} + A\boldsymbol{Z}, \quad \text{where } AA^\top = \Sigma, \tag{66}$$

equivalently,

$$\boldsymbol{X} = \boldsymbol{\mu} + \Sigma^{1/2}\boldsymbol{Z}, \quad \text{where } \Sigma^{1/2}\Sigma^{1/2} = \Sigma \text{ and } \Sigma^{1/2} = \left(\Sigma^{1/2}\right)^T. \tag{67}$$

Motivation for Gaussian Distributions

- For the second motivation, consider modeling random noise in multivariate data (e.g., observational noise such as measurement errors).
- For example, let $X_1, \ldots, X_d$ be the random errors in measuring the $1^{\text{st}}, \ldots, d^{\text{th}}$ coordinates, respectively.
- We may want to design a distribution for such errors satisfying:
  1. The errors in each coordinate are independent, i.e., $X_1, \ldots, X_d$ are mutually independent.
  2. The probability density of an error depends only on its magnitude (distance from the origin), i.e., $f_{\boldsymbol{X}}(\boldsymbol{x})$ is a function of $\boldsymbol{x}^T\boldsymbol{x}$.
- The unique continuous distribution that satisfies the above conditions is the multivariate Gaussian.[6]

[6] This result is known as **Maxwell's Theorem**.

Bivariate Gaussian Distributions

Exercise 30

Let $\boldsymbol{\mu} = (\mu_1, \mu_2)^T$ and $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$ where $(\mu_1, \mu_2)^T \in \mathbb{R}^2$, $\sigma_1 \geq 0$, $\sigma_2 \geq 0$, and $-1 < \rho < 1$. For a given random vector $\boldsymbol{X} = (X_1, X_2)^T$, we denote its Z-score as $\boldsymbol{Z} = (Z_1, Z_2)^T$ where $Z_i = (X_i - \mu_i)/\sigma_i$. Show that the joint PDF of $\boldsymbol{X}$ is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(z_1^2 - 2\rho z_1 z_2 + z_2^2\right)\right). \tag{68}$$

Bivariate Gaussian Distributions

Exercise 31

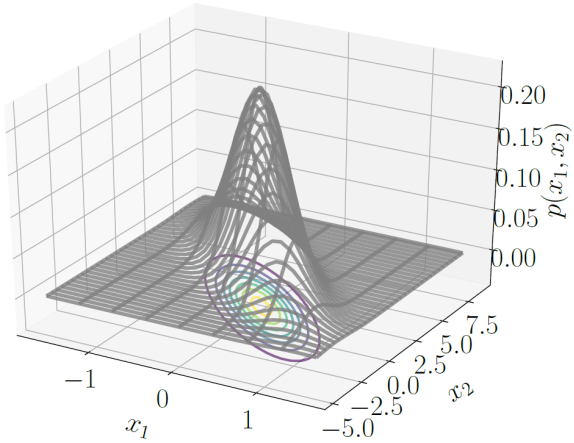
Let Z_1 and Z_2 be i.i.d. standard Gaussian random variables ($\mathcal{N}(0, 1^2)$). Let

$$\begin{aligned} X_1 &= \mu_1 + \sigma_1 Z_1, \\ X_2 &= \mu_2 + \sigma_2 \left(\rho Z_1 + \sqrt{1-\rho^2} Z_2 \right), \end{aligned} \tag{69}$$

where $(\mu_1, \mu_2)^T \in \mathbb{R}^2$, $\sigma_1 \geq 0$, $\sigma_2 \geq 0$, and $-1 < \rho < 1$. Show that

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right). \tag{70}$$

Bivariate Gaussian Distributions



[MML] Fig. 6.7: Plots of bivariate Gaussian PDFs with level curves ($\rho < 0$).

Bivariate Gaussian Distributions

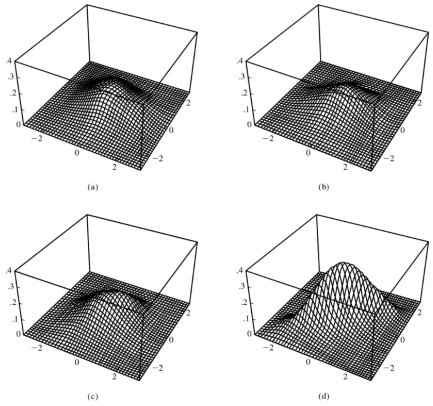


Fig. 3.9. [MSDA]

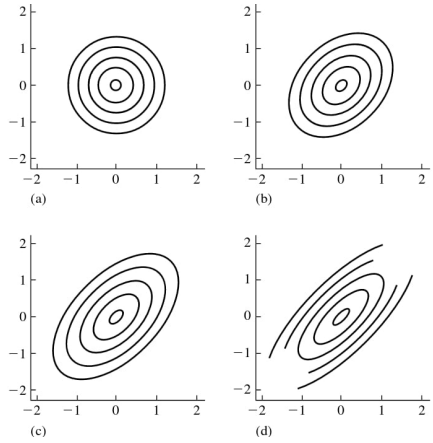


Fig. 3.10. [MSDA]

Marginals of Gaussians are Gaussians

- Marginal distributions of Gaussian random vectors are Gaussians: Let

$$\begin{pmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_{\boldsymbol{X}} \\ \mu_{\boldsymbol{Y}} \end{pmatrix}, \begin{pmatrix} \Sigma_{\boldsymbol{XX}} & \Sigma_{\boldsymbol{XY}} \\ \Sigma_{\boldsymbol{YX}} & \Sigma_{\boldsymbol{YY}} \end{pmatrix} \right). \tag{71}$$

  Then, $\boldsymbol{X} \sim \mathcal{N}(\mu_{\boldsymbol{X}}, \Sigma_{\boldsymbol{XX}})$.

- Note that

$$\begin{pmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \end{pmatrix} = \begin{pmatrix} \mu_{\boldsymbol{X}} \\ \mu_{\boldsymbol{Y}} \end{pmatrix} + \underbrace{\begin{pmatrix} A_{\boldsymbol{XX}} & A_{\boldsymbol{XY}} \\ A_{\boldsymbol{YX}} & A_{\boldsymbol{YY}} \end{pmatrix}}_{A \text{ s.t. } AA^\top = \Sigma} \begin{pmatrix} \boldsymbol{Z}_{\boldsymbol{X}} \\ \boldsymbol{Z}_{\boldsymbol{Y}} \end{pmatrix}, \quad \begin{pmatrix} \boldsymbol{Z}_{\boldsymbol{X}} \\ \boldsymbol{Z}_{\boldsymbol{Y}} \end{pmatrix} \sim \mathcal{N}(0_k, \mathrm{I}_k). \tag{72}$$

  Therefore, $\boldsymbol{X} = \mu_{\boldsymbol{X}} + \begin{pmatrix} A_{\boldsymbol{XX}} & A_{\boldsymbol{XY}} \end{pmatrix} \begin{pmatrix} \boldsymbol{Z}_{\boldsymbol{X}} \\ \boldsymbol{Z}_{\boldsymbol{Y}} \end{pmatrix}$ and $\begin{pmatrix} A_{\boldsymbol{XX}} & A_{\boldsymbol{XY}} \end{pmatrix} \begin{pmatrix} A_{\boldsymbol{XX}}^T \\ A_{\boldsymbol{XY}}^T \end{pmatrix} = \Sigma_{\boldsymbol{XX}}$.

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